

FIELDCELL · REQUIREMENTS

FieldCell requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldcell		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Status of every requirement from FCL-CAL-001; R2 names 120 V 60 Hz for the first build (FCL-DDR-001); irradiance basis stated
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

FieldCell requirements

These are the requirements for the concept. The value and status columns come from the TRL 3 calculation note FCL-CAL-001, which checks each one by calculation. Six are met, two are not met (R6 mass and R12 cost), two are at risk (R7 and R9) and two cannot be verified until test (R8 and R11). No target has been relaxed; proposals to change R6 and the budget behind R12 await Amish (docs/REVIEW.md).

Table 1. Requirements.

ID	Requirement	Target	Value at TRL 3 (FCL-CAL-001)	Status	Verification
R1	Store enough energy for a day of field use	1.2 kWh nominal or more; 1.0 kWh usable or more	1.28 kWh nominal, 1.15 kWh usable at 90 % depth of discharge	Met	Calculation; datasheet
R2	Supply AC and DC loads	1,000 W continuous pure sine AC at 120 V 60 Hz for the first build (230 V 50 Hz variant), 2,000 W surge; DC outlets 300 W or more in total (USB-C PD and 12 V)	1 kW inverter by selection; 440 W DC; 66 A continuous battery current on a 100 A fuse and BMS	Met	Calculation; datasheets
R3	Recharge from its own PV	1.1 kWh/day or more into the battery at 4 peak sun hours; full recharge from 10 % in one day at 5 peak sun hours with no load	1.17 kWh/day at 4 kWh/m ² /day (1.09 in 40 °C air); full recharge in 0.79 day at 5	Met	PV yield calculation; later logged trial
R4	Carry a reference load through a day without sun	Reference load of 0.9 kWh/day at the outlets for 1 day or more with no sun; energy-neutral at 4 peak sun hours or more	1.08 days; +10 % surplus at 4 kWh/m ² /day; break-even at 3.64	Met	Load and energy balance; later logged trial
R5	Deploy fast with one person	From arrival to power available in 10 min or less, one person, no tools; stow in 10 min or less	6.0 min deploy, 5.0 min stow by task analysis	Met	Task analysis; later timed trial
R6	Movable by one person	Total mass 70 kg or less; vertical handle force 10 to 150 N (never negative) over a 5° pitch either way; heaviest removable module 25 kg or less	73.5 kg; 14 to 50 N; pack 12.0 kg	Not met (mass 3.5 kg over)	Mass budget and CG calculation; later weighing
R7	Cross rough ground	16 in (406 mm) wheels or larger; ground clearance 150 mm or more; climb a 150 mm step; pull force 150 N or less on a 10 % grade over gravel or grass; static lateral tip angle 25° or more	406 mm wheels; 193 mm clearance; step with about 136 N pull at the grip; 115 N gravel, 144 N grass; 32.9°	At risk (grass pull within 4 % of limit)	Geometry and force calculation
R8	Work in rain and dust	Electronics enclosure IP54 while operating; battery enclosure IP65; all cable entries through glands or sealed connectors	IP54 enclosure with IP54 filter fan and exhaust filter; IP65 case; glands, by selection	Not verifiable at TRL 3	Design review; later spray and dust test
R9	Stay safe across field temperatures	Operate at –10 to 45 °C ambient; BMS blocks charging below 0 °C cell temperature; inverter derating known at 45 °C	Charge block by BMS selection; electronics box about 54 °C at 45 °C ambient with the fan (70 °C sealed); pack in sun about 55 °C, so charging is blocked in the hottest hours	At risk	Thermal estimate; datasheets

R10

ID	Requirement	Target	Value at TRL 3 (FCL-CAL-001)	Status	Verification
	Stay put in wind when deployed	Wings and cart stable to 15 m/s (54 km/h) with the four outrigger feet staked or ballasted; stow instruction above that	Staked: 27 N pull-out per foot at 15 m/s (49 N with a 1.5 factor); unstaked wings lift at about 9.2 m/s	Met (staked)	Wind load calculation
R11	Be quiet	45 dB(A) or less at 1 m at 500 W load	About 41 dB(A) from assumed fan levels, not datasheet values	Not verifiable at TRL 3	Datasheets; later sound level measurement
R12	Be electrically safe and affordable	Every circuit fused at its source; GFCI (120 V) or 30 mA RCD (230 V) on all AC outlets; battery disconnect reachable without tools; no inlet or path to building wiring; parts cost \$1,500 or less	Safety provisions met by design; parts about \$2,006	Not met (cost 34 % over)	Design review; priced BOM

■ Assumptions

- "Peak sun hours" in R3 and R4 is read as daily global horizontal irradiation in kWh/m²/day, the figure most climate data give. FCL-CAL-001 converts it to the east-west wings with a transposition ratio of about 0.95. This reading is proposed, awaiting Amish.
- 4 to 5 kWh/m²/day covers clear-season sites at low and middle latitudes; a cloudy week or a high-latitude winter is not covered by R3 or R4.
- The reference load in R4 is defined in FCL-PRC-001, Table 3: lighting, radio and phone charging, a laptop and router, and short power-tool bursts.
- Usable battery capacity is 90 % of nominal, for LiFePO₄ cells with a conservative BMS.
- "Arrival" in R5 means the cart has been wheeled to the chosen spot; choosing the spot is not timed.
- The budget in R12 is `budget_usd` in `project.yaml` (\$1,500), kept by Amish's decision of 2026-09-25 pending supplier quotes (FCL-DDR-001, D7).